Advanced Calculus I Hand-in Assignment 6

Student ID Number (學號): Name :

- This Assignment **will be collected in class.** (這份作業會在上課的時候收。) Please put your assignment on the desk in our classroom **before 10:25 on Oct 28th, Wednesday**. (請在10月28日10:25分以前，將作業放在教室的講台。) Please write down your student ID number, name and answer on A4 papers and staple your papers together （請將學號、姓名、答案寫在A4大小的紙張。若有兩張以上的答案紙，請用訂書機訂好。).
- Note that we do not collect late assignment. (請注意，不收遲交作業。)
- 作業在第2頁附上中文，希望可以在學習英文上對同學有所幫助，但是小考、期中、期末考不會附上中文，**建議同學先讀第一頁全英文的部份，再比較理解的情況和第二頁有什麼不同**。寫作業、考試可以寫中文或是中文、英文混合。

Problem 1. (10 points.) Let $E = \{\frac{1}{n} : n \in \mathbb{N}\}$. Please determine whether E is compact or not. Note that please give explanation.

Problem 2. (10 points) Suppose that $S \subset \mathbb{R}$ be nonempty and bounded above. Please construct a sequence of nested closed intervals

$$I_n = [x_n, y_n]$$

satisfying

1. x_n or $y_n \in S$, for all $n \in \mathbb{N}$,
2. y_n is an upper bound of S, for all $n \in \mathbb{N}$,
3. $|I_n| \to 0$ as $n \to \infty$,
4. Sup $S = \bigcap_{n=1}^{\infty} I_n$

Bonus Problem. (3 points) Let E be sebset in $\mathbb{R}^n$. Please write down the definition of the following terminology.
(1) An *open covering* of E. (1 point)
(2) E is *compact*. (2 points)

Problem 1. (10 分.)令$E = \{\frac{1}{n} : n \in \mathbb{N}\}$。請判斷$E$是否爲緊緻集。請注意，請附上解釋。

Problem 2. (10 分) 令S爲實數空間中的非空子集合而且有上界。請造出一個區間套

$$I_n = [x_n, y_n]$$

，該區間套同時滿足下列四個條件：

1. 對於每一個正整數n，x_n或是y_n在集合S中,
2. 對於每一個正整數n，y_n是集合S的上界，
3. 當$n \to \infty$時，區間長度，$|I_n| \to 0$，
4. Sup $S = \bigcap_{n=1}^{\infty} I_n$ (集合S的最小上界會是該區間套的交集。)

Bonus Problem. (3 points) 令E爲$\mathbb{R}^n$空間中的集合。請寫出下列名詞的定義。
(1) E的開覆蓋。(1 分)
(2) E爲緊緻集。 (2 分)